



LAB 5 REPORT

CYOP 300 - BUILDING SECURE WEB APPLICATIONS

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Executive Summary

Attached is my submission for our Lab5 activity.

As with previous labs, I made use of 3rd party Python modules to create a text-based UI for application navigation. This is helpful because it prevents the user from entering any unexpected input during functionality selection, eliminating a potential source of bugs. Unlike previous submissions, this lab has made use of a module that fully supports running in both the Windows “Command Prompt” shell, running cmd.exe, as well as the Terminal shell, running Powershell.

Of note is that this lab’s application is split across three python modules, main.py, load_data.py, and lab5ui.py. My original intent was to separate the UI from data processing components, however, after more consideration of the project instructions, I believed that it would introduce unnecessary complexity.

Full unit tests were generated and are included in the project zip file.

Pylint results are included at the end of this report.

Testing

Test Case Table

Test Cases	Description	Expected Output	Actual Output	Pass/Fail
Test Case 1	Demonstrate handling of missing CSV files: Rename Housing.csv to a temporary name, execute the program, and then re-name the temporary name back to Housing.csv	Message: Both Housing.csv and PopChange.csv are required. Please ensure they are present in <i>current working directory</i> .	Message: Both Housing.csv and PopChange.csv are required. Please ensure they are present in <i>current working directory</i> .	Pass
Test Case 2	Demonstrate exiting the application: Select 'Exit' from the top menu	Select the file you want to analyze: Exit ***** Thanks for using the Data Analysis App *****	Select the file you want to analyze: Exit ***** Thanks for using the Data Analysis App *****	Pass
Test Case 3	Demonstrate exiting a submenu and returning to the main menu: Select Population Data, then select 'd. Exit'	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: d. Exit Exiting Population Data analysis. ? Select the file you want to analyze: Population Data Housing Data Exit	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: d. Exit Exiting Population Data analysis. ? Select the file you want to analyze: > Population Data	Pass

			Housing Data	
			Exit	
Test Case 4	Show Population Data – Pop April 1 data and histogram	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed.	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed.	Pass
Test Case 5	Show Population Data – Pop April 1 data and histogram <u>without</u> returning to the main menu	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1	Pass

		You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze:	You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed.	
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		> a. Pop Apr 1 b. Pop Jul 1 c. Change Pop d. Exit	You have entered Population Data. ? Select the Column you want to analyze: > a. Pop Apr 1 b. Pop Jul 1 c. Change Pop d. Exit	
Test Case 6	Show Population Data – Pop July 1 data and histogram	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed.	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0	Pass

		The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 b. Pop Jul 1 c. Change Pop d. Exit	The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed. The Histogram of this column is now displayed. You have entered Population Data. ? Select the Column you want to analyze: > a. Pop Apr 1 b. Pop Jul 1	
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			c. Change Pop d. Exit	
Test Case 7	Show Population Data – Change Pop data and histogram	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: c. Change Pop You selected: Change Pop The statistics for this column are: Count = 556 Mean = 154.8 Standard Deviation = 3050.6 Min = -13775.0 Max = 22363.0 The Histogram of Change Pop is now displayed.	? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: c. Change Pop You selected: Change Pop The statistics for this column are: Count = 556 Mean = 154.8 Standard Deviation = 3050.6 Min = -13775.0 Max = 22363.0 The Histogram of Change Pop is now displayed.	Pass
Test Case 8	Show Housing Data - Age	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: a. AGE	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: a. AGE	Pass

		You selected: AGE The statistics for this column are: Count = 10042 Mean = 47.2 Standard Deviation = 23.1 Min = -9.0 Max = 93.0 The Histogram of AGE is now displayed.	You selected: AGE The statistics for this column are: Count = 10042 Mean = 47.2 Standard Deviation = 23.1 Min = -9.0 Max = 93.0 The Histogram of AGE is now displayed.	
Test Case 9	Show Housing Data - Bedrooms	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: b. BEDRMS You selected: BEDRMS The statistics for this column are: Count = 10042 Mean = 2.7 Standard Deviation = 1.1 Min = 0.0 Max = 7.0 The Histogram of BEDRMS is now displayed.	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: b. BEDRMS You selected: BEDRMS The statistics for this column are: Count = 10042 Mean = 2.7 Standard Deviation = 1.1 Min = 0.0 Max = 7.0 The Histogram of BEDRMS is now displayed.	Pass
Test Case 10	Show Housing Data - Built	? Select the file you want to analyze: Housing Data	? Select the file you want to analyze: Housing Data	Pass

		You have entered Housing Data. ? Select the Column you want to analyze: c. BUILT You selected: BUILT The statistics for this column are: Count = 10042 Mean = 1967.0 Standard Deviation = 26.3 Min = 1919.0 Max = 2012.0 The Histogram of BUILT is now displayed.	You have entered Housing Data. ? Select the Column you want to analyze: c. BUILT You selected: BUILT The statistics for this column are: Count = 10042 Mean = 1967.0 Standard Deviation = 26.3 Min = 1919.0 Max = 2012.0 The Histogram of BUILT is now displayed.	
Test Case 11	Show Housing Data - Rooms	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: d. ROOMS You selected: ROOMS The statistics for this column are: Count = 10042 Mean = 5.7 Standard Deviation = 1.9 Min = 1.0 Max = 14.0	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: d. ROOMS You selected: ROOMS The statistics for this column are: Count = 10042 Mean = 5.7 Standard Deviation = 1.9 Min = 1.0 Max = 14.0	Pass

		The Histogram of ROOMS is now displayed. You have entered Housing Data.	The Histogram of ROOMS is now displayed. You have entered Housing Data.	
Test Case 12	Show Housing Data - Utility	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: e. UTILITY You selected: UTILITY The statistics for this column are: Count = 10042 Mean = 189.6 Standard Deviation = 128.9 Min = 0.0 Max = 1107.6 The Histogram of UTILITY is now displayed.	? Select the file you want to analyze: Housing Data You have entered Housing Data. ? Select the Column you want to analyze: e. UTILITY You selected: UTILITY The statistics for this column are: Count = 10042 Mean = 189.6 Standard Deviation = 128.9 Min = 0.0 Max = 1107.6 The Histogram of UTILITY is now displayed.	Pass
Test Case 13	Runs on Windows without WSL	Run program, select any menu options required to generate a histogram.	<pre>(cyop300) PS Z:\> cmd.exe Microsoft Windows [Version 10.0.26100.8037] (c) Microsoft Corporation. All rights reserved. Z:\>exit (cyop300) PS Z:\> python main.py</pre>	Pass

			***** Welcome to the Python Data Analysis App ***** ? Select the file you want to analyze: Population Data You have entered Population Data. ? Select the Column you want to analyze: a. Pop Apr 1 You selected: Pop Apr 1 The statistics for this column are: Count = 556 Mean = 49957.3 Standard Deviation = 27250.7 Min = 13519.0 Max = 218478.0 The Histogram of Pop Apr 1 is now displayed.	
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Test Case Screenshots

Test Case 1

```
^ mv Housing.csv tempfile

alberth .../CYOP300/lab5 lab5 X!+? v3.14.4(#.venv) 7:58 PM
^ python main.py
Both Housing.csv and PopChange.csv are required. Please ensure they are present in /Users/alberth/src/CYOP300/lab5.

alberth .../CYOP300/lab5 lab5 X!+? v3.14.4(#.venv) 7:58 PM
^ mv tempfile Housing.csv
```

Test Case 2

```
^ python main.py
***** Welcome to the Python Data Analysis App *****
? Select the file you want to analyze: Exit

***** Thanks for using the Data Analysis App *****
```

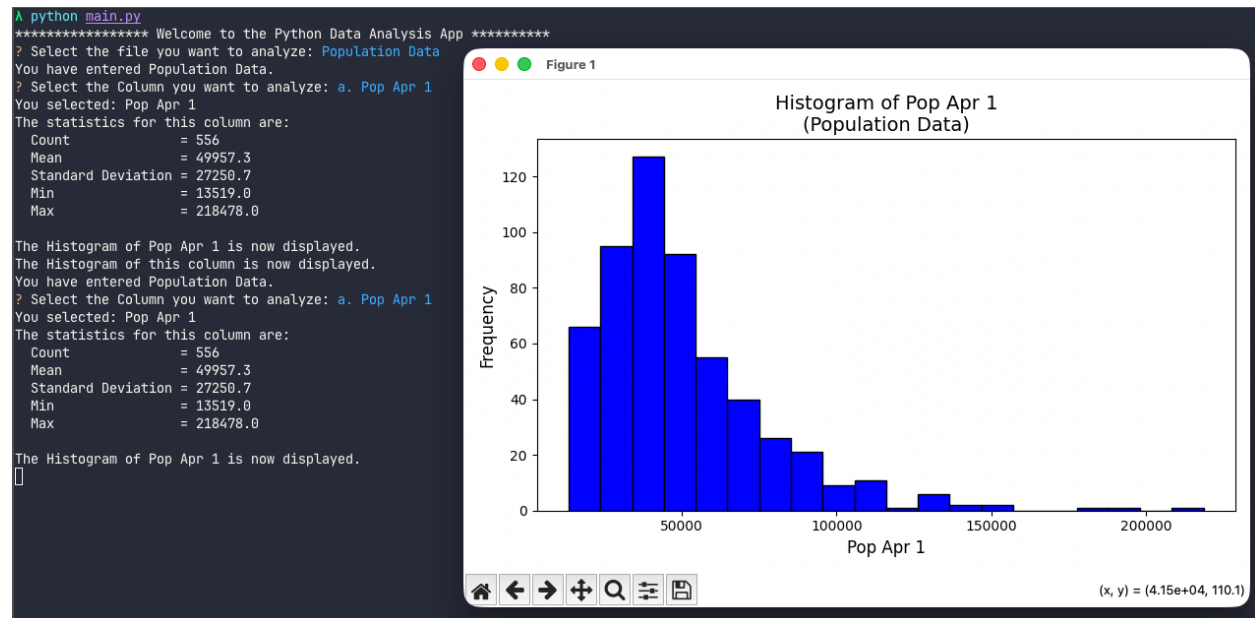
Test Case 3

```
***** Welcome to the Python Data Analysis App *****
? Select the file you want to analyze: Population Data
You have entered Population Data.
? Select the Column you want to analyze: d. Exit
Exiting Population Data analysis.
? Select the file you want to analyze: 
) Population Data
  Housing Data
  Exit
```

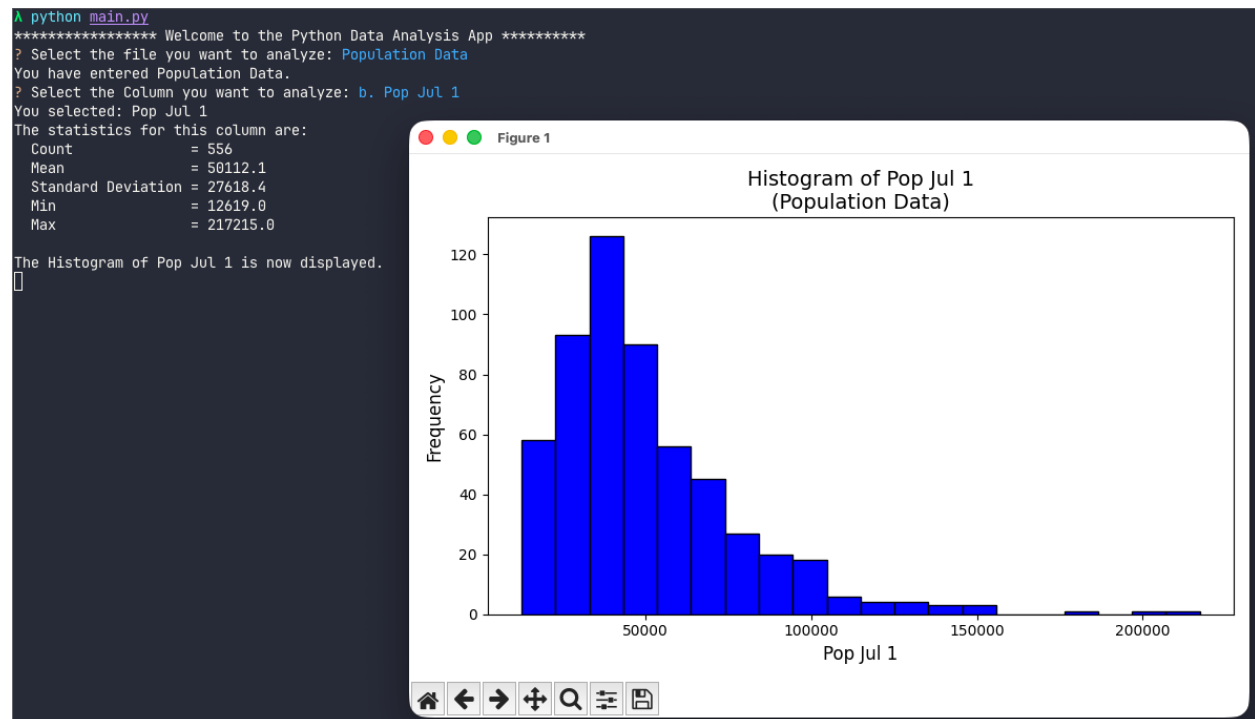
Test Case 4

```
***** Welcome to the Python Data Analysis App *****
? Select the file you want to analyze: Population Data
You have entered Population Data.
? Select the Column you want to analyze: d. Exit
Exiting Population Data analysis.
? Select the file you want to analyze: 
) Population Data
  Housing Data
  Exit
```

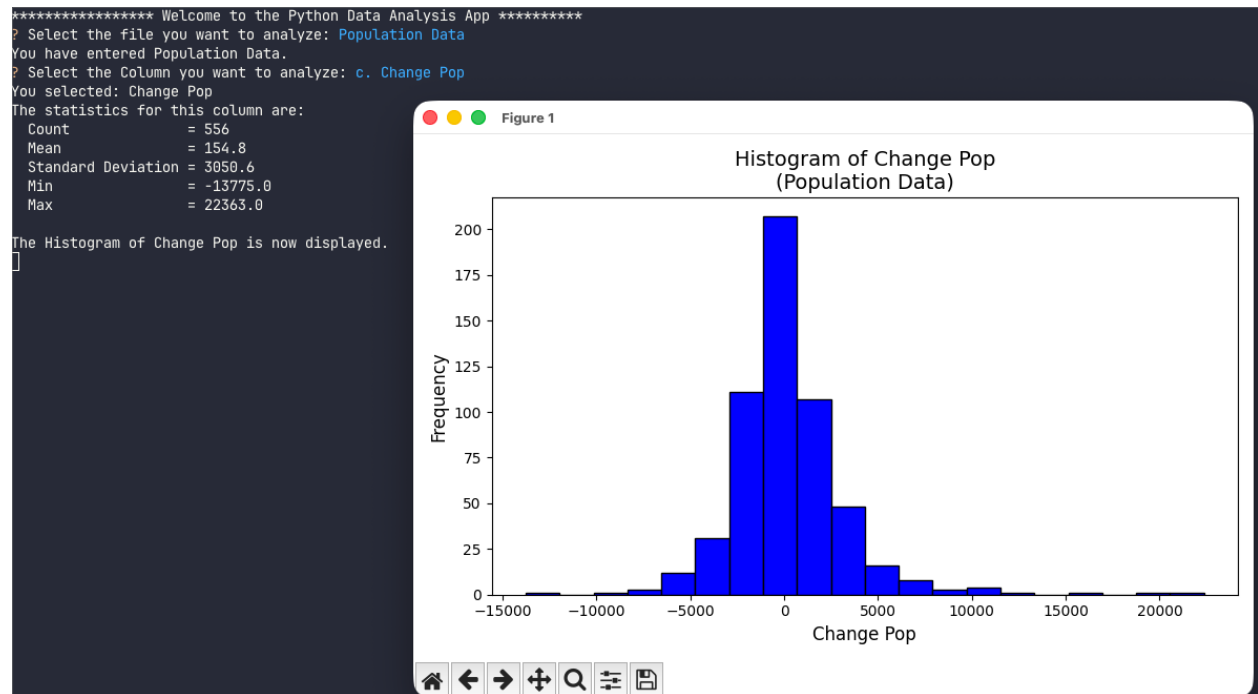
Test Case 5



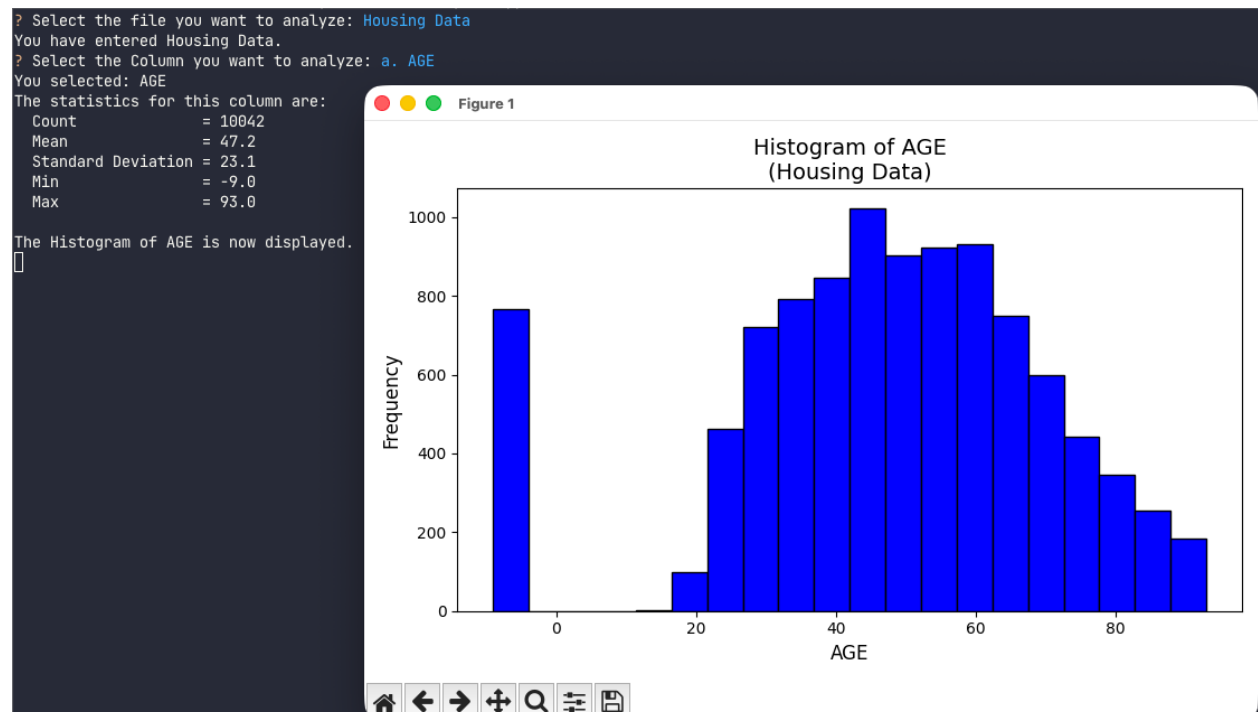
Test Case 6



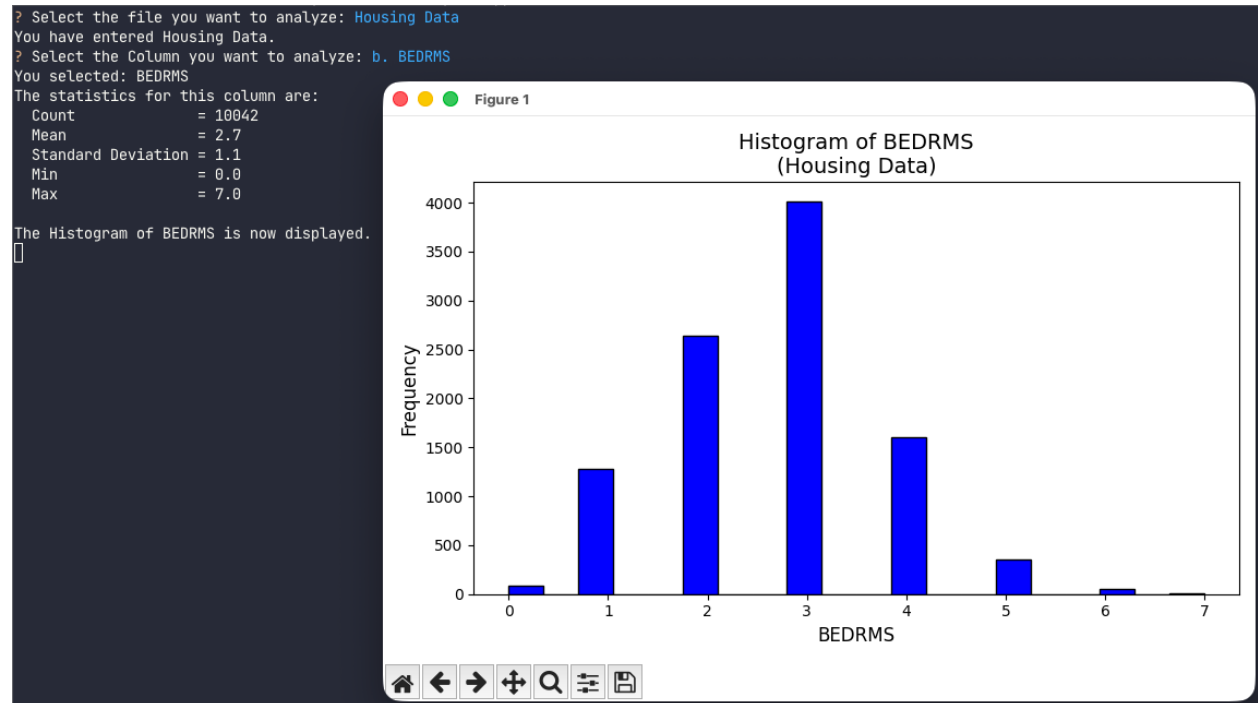
Test Case 7



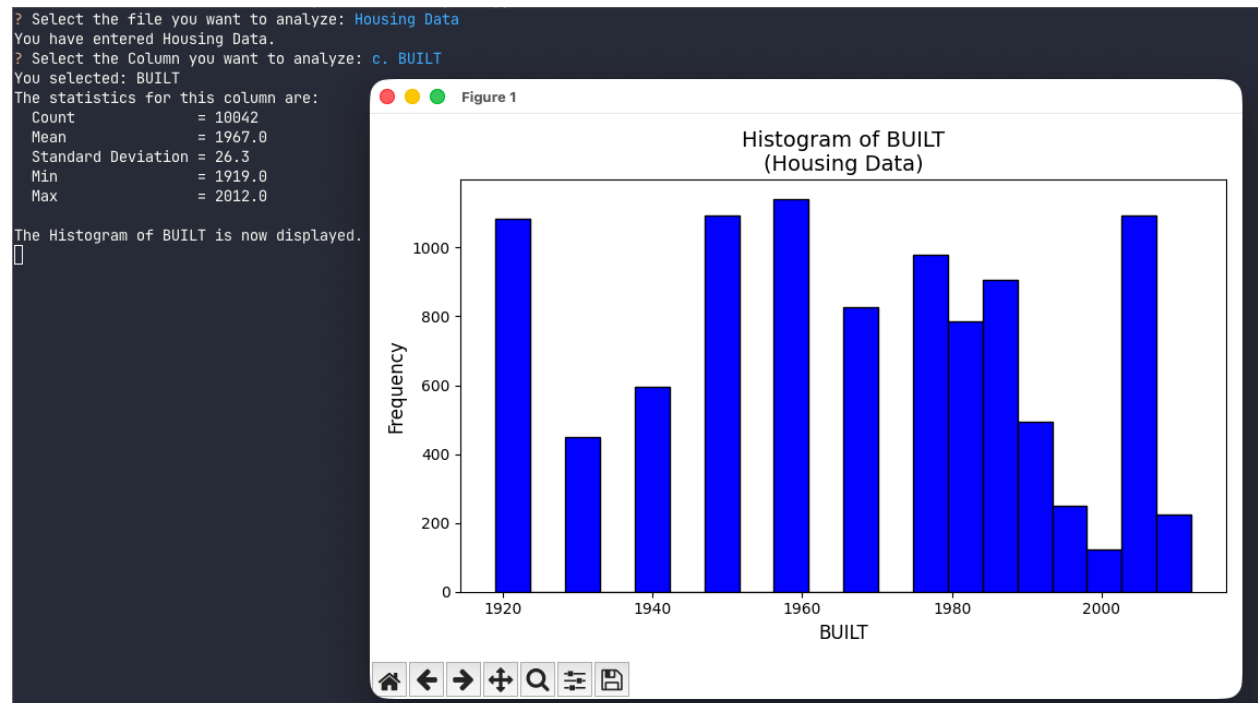
Test Case 8



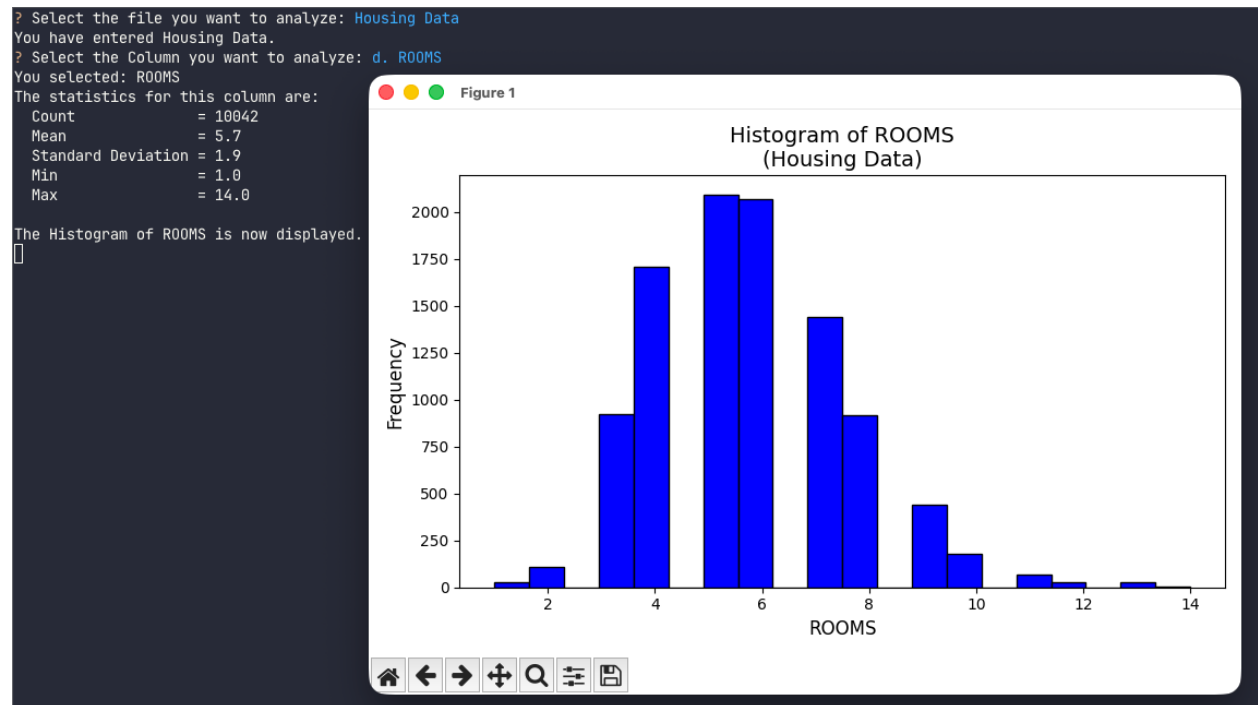
Test Case 9



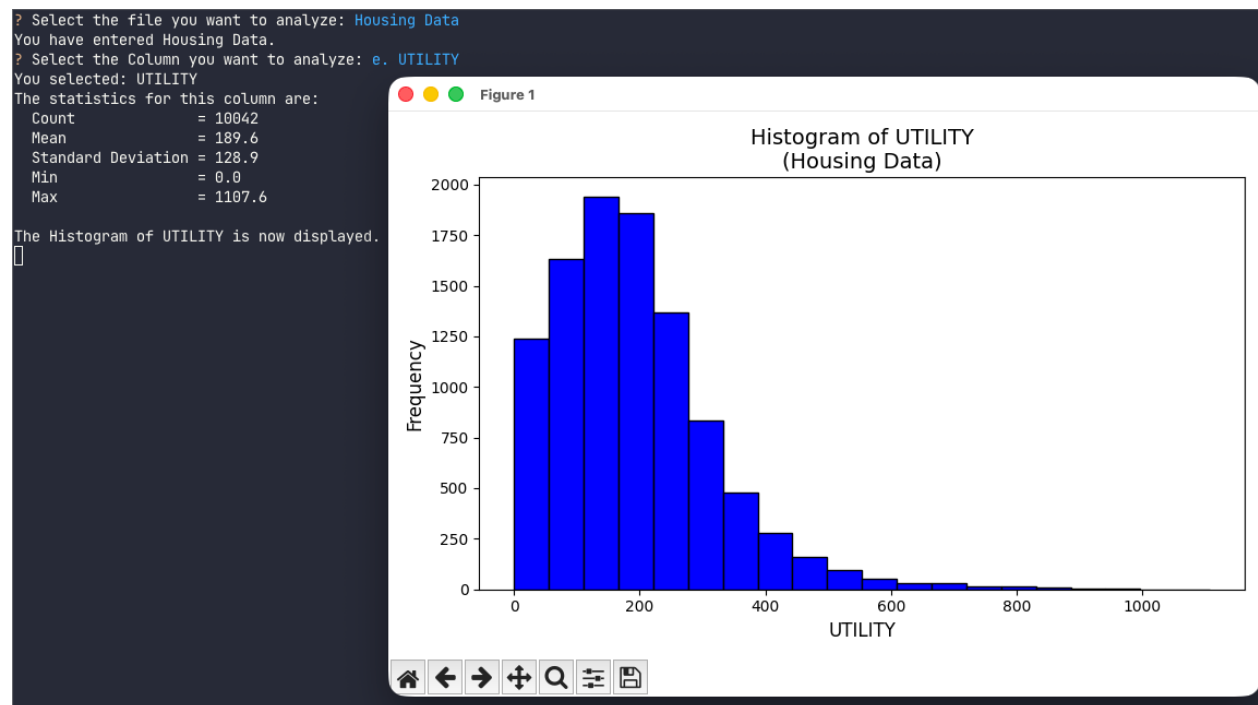
Test Case 10



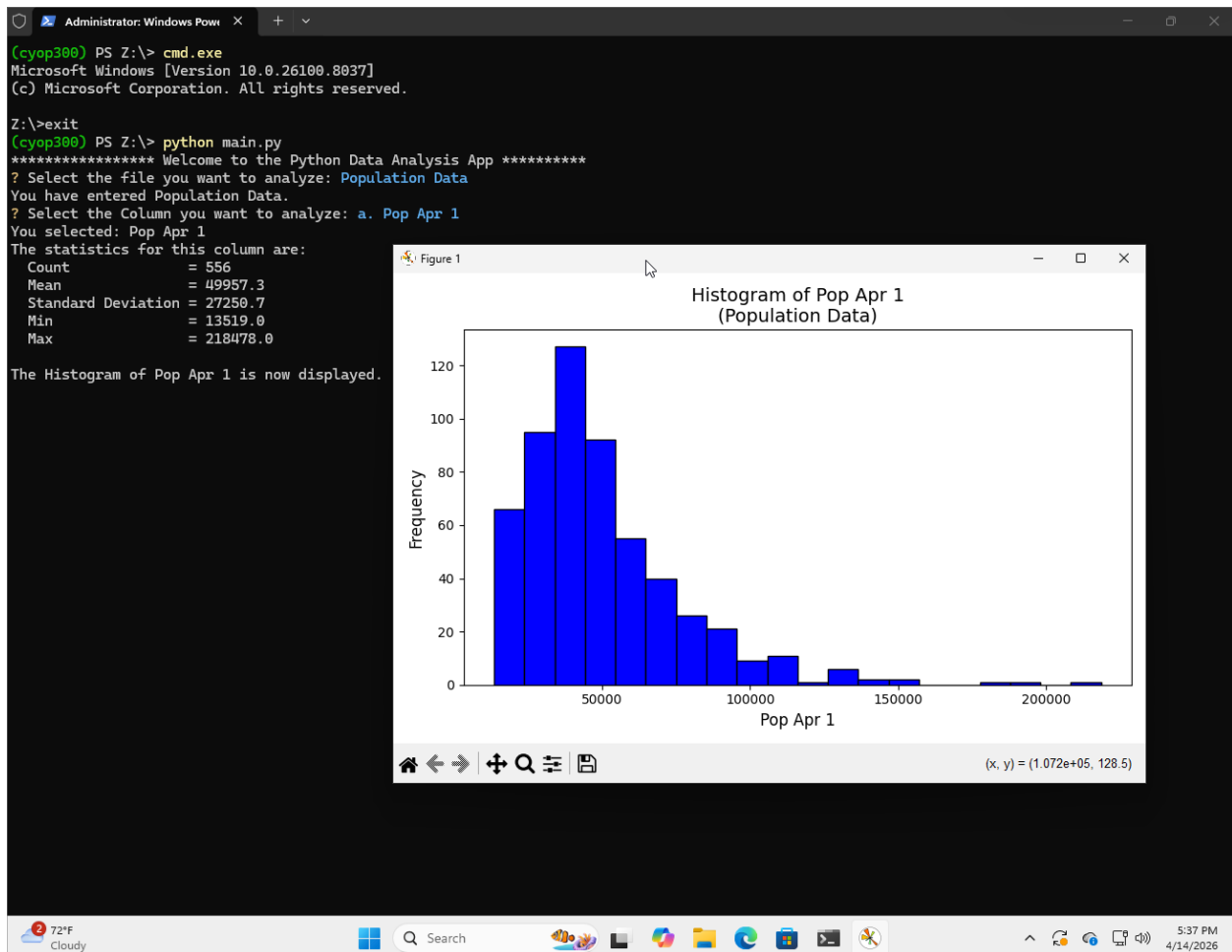
Test Case 11



Test Case 12



Test Case 12



Pylint Results

```
λ pylint *.py
```

Your code has been rated at 10.00/10